Approximate methods

In the preceeding chapters, the methods to find the natural frequencies and mode shapes of multi degree freedom systems were presented. It is done by setting the characteristic determinant equal to zero and then solving the resulting equation. If the number of degrees of freedom is high, the solution becomes very tedious. More over, we may not be interested in knowing all the natural frequencies of the given system. Here, the proceedure for determining the approximate fundamental frequency of a multi-degree freedom system is presented.

1. Dunkerley's Method

Consider an n-degree of freedom system. Its natural frequencies (eigen values) are found out by solving the equation $\det\{[k]-\omega^2[m]\}=0$

$$|-[k]+\omega^2[m]|=0$$

Multiplying by $[k]^{-1}$

$$|-[k]^{-1}[k]+\omega^2[k]^{-1}[m]|=0$$

$$\left|-[I]+\omega^2[a][m]\right|=0$$

Where $[k]^{-1}[k]=[I]$ and $[k]^{-1}=[a]$, the flexibility influence coefficient matrix.

$$\omega^2\left|-\frac{1}{\omega^2}[I]+[a][m]\right|=0 \implies\implies \left|-\frac{1}{\omega^2}[I]+[a][m]\right|=0$$

For a lumped mass system with diagonal mass matrix

$$\left|-\frac{1}{\omega^2}\begin{pmatrix}1 & 0 & \dots & 0\\ 0 & 1 & \dots & 0\\ \vdots & \vdots & & \vdots\\ 0 & & \dots & 1\end{pmatrix}+\begin{bmatrix}a_{11} & a_{12} & \cdots & a_{1n}\\ a_{21} & a_{22} & \cdots & a_{2n}\\ \vdots & \vdots & & \vdots\\ a_{n1} & a_{n2} & \cdots & a_{nn}\end{bmatrix}\begin{bmatrix}m_1 & 0 & \dots & 0\\ 0 & m_2 & \dots & 0\\ \vdots & \vdots & & \vdots\\ 0 & 0 & \dots & m_n\end{bmatrix}\right|=0$$

$$\left|\begin{pmatrix}-\frac{1}{\omega^2} & 0 & \dots & 0\\ 0 & -\frac{1}{\omega^2} & \dots & 0\\ \vdots & \vdots & & \vdots\\ 0 & & \dots & -\frac{1}{\omega^2}\end{pmatrix}+\begin{bmatrix}a_{11}m_1 & a_{12}m_2 & \cdots & a_{1n}m_n\\ a_{21}m_1 & a_{22}m_2 & \cdots & a_{2n}m_n\\ \vdots & \vdots & & \vdots\\ a_{n1}m_1 & a_{n2}m_2 & \cdots & a_{nn}m_n\end{bmatrix}\right|=0$$

$$\begin{vmatrix}\left(-\frac{1}{\omega^2}+a_{11}m_1\right) & a_{12}m_2 & \dots & a_{1n}m_n\\ a_{21}m_1 & \left(-\frac{1}{\omega^2}+a_{22}m_2\right) & \dots & a_{2n}m_n\\ \vdots & \vdots & & \vdots\\ a_{n1}m_1 & a_{n2}m_2 & \dots & \left(-\frac{1}{\omega^2}+a_{nn}m_n\right)\end{vmatrix}=0$$

Let us assume for the time being that $n=3$

$$\begin{vmatrix}\left(-\frac{1}{\omega^2}+a_{11}m_1\right) & a_{12}m_2 & a_{13}m_3\\ a_{21}m_1 & \left(-\frac{1}{\omega^2}+a_{22}m_2\right) & a_{23}m_3\\ a_{31}m_1 & a_{32}m_2 & \left(-\frac{1}{\omega^2}+a_{33}m_3\right)\end{vmatrix}=0$$

$$\left(-\frac{1}{\omega^2}+a_{11}m_1\right)\left(-\frac{1}{\omega^2}+a_{22}m_2\right)\left(-\frac{1}{\omega^2}+a_{33}m_3\right)+a_{13}m_3a_{21}m_1a_{32}m_2+a_{12}m_2a_{23}m_3a_{31}m_1-a_{31}m_1\left(-\frac{1}{\omega^2}+a_{22}m_2\right)a_{13}m_3 -a_{21}m_1a_{12}m_2\left(-\frac{1}{\omega^2}+a_{33}m_3\right)-a_{32}m_2a_{23}m_3\left(-\frac{1}{\omega^2}+a_{11}m_1\right)=0$$

$$\left(-\frac{1}{\omega^2}\right)^3+\left(-\frac{1}{\omega^2}\right)^2a_{22}m_2+\left(-\frac{1}{\omega^2}\right)^2a_{33}m_3+\left(-\frac{1}{\omega^2}\right)^2a_{11}m_1+\left(-\frac{1}{\omega^2}\right)a_{22}m_2a_{33}m_3+\left(-\frac{1}{\omega^2}\right)a_{11}m_1a_{33}m_3 +\left(-\frac{1}{\omega^2}\right)a_{11}m_1a_{22}m_2+a_{11}m_1a_{22}m_2a_{33}m_3=0$$

$$\left(-\frac{1}{\omega^2}\right)^3+\left(-\frac{1}{\omega^2}\right)^2(a_{11}m_1+a_{22}m_2+a_{33}m_3)+\left(-\frac{1}{\omega^2}\right)(a_{11}m_1a_{22}m_2+a_{22}m_2a_{33}m_3+a_{11}m_1a_{33}m_3)+a_{11}m_1a_{22}m_2a_{33}m_3=0---(1)$$

This is a polynomial of degree 3 in $\left(-\frac{1}{\omega^2}\right)$. Let the roots of this polynomial is $\left(-\frac{1}{{\omega_1}^2}\right)$, $\left(-\frac{1}{{\omega_2}^2}\right)$, and $\left(-\frac{1}{{\omega_3}^2}\right)$. Then,

$$\left\{\left(-\frac{1}{\omega^2}\right)-\left(-\frac{1}{{\omega_1}^2}\right)\right\}\left\{\left(-\frac{1}{\omega^2}\right)-\left(-\frac{1}{{\omega_2}^2}\right)\right\}\left\{\left(-\frac{1}{\omega^2}\right)-\left(-\frac{1}{{\omega_3}^2}\right)\right\}=0$$

$$\left(-\frac{1}{\omega^2}\right)^3-\left(-\frac{1}{\omega^2}\right)^2\left(-\frac{1}{{\omega_1}^2}-\frac{1}{{\omega_2}^2}-\frac{1}{{\omega_3}^2}\right)+\left(-\frac{1}{\omega^2}\right)\left(\frac{-1}{{\omega_1}^2}\times\frac{-1}{{\omega_2}^2}+\frac{-1}{{\omega_2}^2}\times\frac{-1}{{\omega_3}^2}+\frac{-1}{{\omega_1}^2}\times\frac{-1}{{\omega_3}^2}\right)-\left(\frac{-1}{{\omega_1}^2}\times\frac{-1}{{\omega_2}^2}\times\frac{-1}{{\omega_3}^2}\right)=0---(2)$$

Comparing equations (1) and (2), we can write

$$-\left(-\frac{1}{\omega^2}\right)^2\left(-\frac{1}{{\omega_1}^2}-\frac{1}{{\omega_2}^2}-\frac{1}{{\omega_3}^2}\right)=\left(-\frac{1}{\omega^2}\right)^2(a_{11}m_1+a_{22}m_2+a_{33}m_3)$$

$$\frac{1}{{\omega_1}^2}+\frac{1}{{\omega_2}^2}+\frac{1}{{\omega_3}^2}=a_{11}m_1+a_{22}m_2+a_{33}m_3$$

Extending this result to n-degree-of-freedom system

$$\frac{1}{{\omega_1}^2}+\frac{1}{{\omega_2}^2}+\frac{1}{{\omega_3}^2}+\cdots\ldots\ldots\ldots..+\frac{1}{{\omega_n}^2}=a_{11}m_1+a_{22}m_2+a_{33}m_3+\cdots\ldots\ldots\ldots\ldots+a_{nn}m_n---(3)$$

In general, higher natural frequencies ω_2, ω_3, ...ω_n are considerably larger than ω_1, $\frac{1}{{\omega_2}^2}$, $\frac{1}{{\omega_3}^2}$, ..., $\frac{1}{{\omega_n}^2}$ will be very small compared with $\frac{1}{{\omega_1}^2}$.

$$\therefore (3) \Longrightarrow\Longrightarrow \boldsymbol{\frac{1}{{\omega_1}^2}\simeq a_{11}m_1+a_{22}m_2+a_{33}m_3}$$

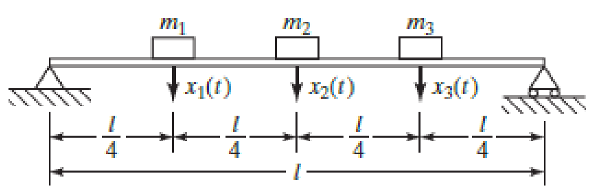
Extending this to n-degree-of-freedom

$$\therefore (3) \Longrightarrow\Longrightarrow \boldsymbol{\frac{1}{{\omega_1}^2}\simeq a_{11}m_1+a_{22}m_2+a_{33}m_3+\cdots\ldots\ldots\ldots\ldots+a_{nn}m_n---(4)}$$

This is known as Dunkerley's formula. The fundamental frequency given by this formula will be aways lower than the exact value.

Example 1

Determine the fundamental frequency of the simply supported beam using Dunkerly's formula. Assume $m_1 = m_2 = m_3$.



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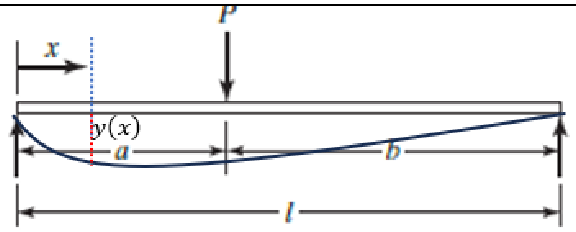
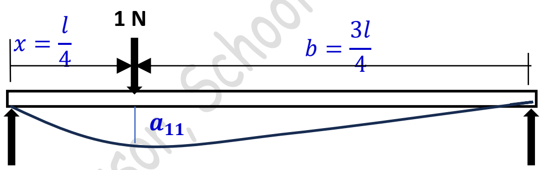


The reader may refer to the topics of Mechanics of Solids and recall the equation for the deflection of a simply supported beam at a location x from one support when a load P is acting at a distance b from the other support given by

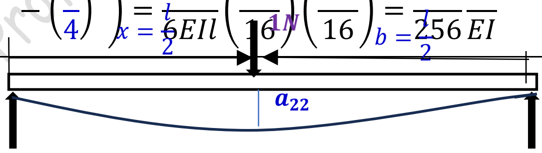
$$y(x) = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2); \quad 0 \le x \le a$$

Flexibility influence coefficient $\boldsymbol{a_{11}}$ is the deflection at location 1 due to a unit load at location 1.



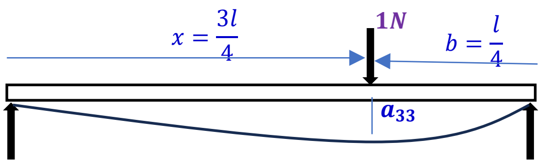
$$a_{11} = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2) = \frac{(1N)\left(\frac{3l}{4}\right)\left(\frac{l}{4}\right)}{6EIl}\left(l^2 - \left(\frac{3l}{4}\right)^2 - \left(\frac{l}{4}\right)^2\right) = \frac{1}{6EIl}\left(\frac{3l^2}{16}\right)\left(\frac{6l^2}{16}\right) = \frac{3}{256}\frac{l^3}{EI}$$

Flexibility influence coefficient $\boldsymbol{a_{22}}$ is the deflection at location 2 due to a unit load at location 2.



$$a_{22} = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2) = \frac{(1N)\left(\frac{l}{2}\right)\left(\frac{l}{2}\right)}{6EIl}\left(l^2 - \left(\frac{l}{2}\right)^2 - \left(\frac{l}{2}\right)^2\right) = \frac{(m_2 g)}{6EIl}\left(\frac{l^2}{4}\right)\left(\frac{l^2}{2}\right) = \frac{1}{48}\frac{l^3}{EI}$$

Flexibility influence coefficient $\boldsymbol{a_{33}}$ is the deflection at location 3 due to a unit load at location 3.



$$a_{33} = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2) = \frac{(1N)\left(\frac{l}{4}\right)\left(\frac{3l}{4}\right)}{6EIl}\left(l^2 - \left(\frac{l}{4}\right)^2 - \left(\frac{3l}{4}\right)^2\right) = \frac{(m_{31} g)}{6EIl}\left(\frac{3l^2}{16}\right)\left(\frac{6l^2}{16}\right) = \frac{3}{256}\frac{l^3}{EI}$$

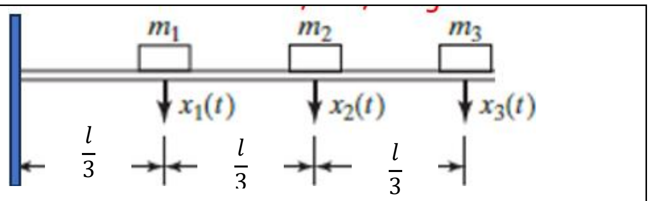
From Dunkerley's formula,

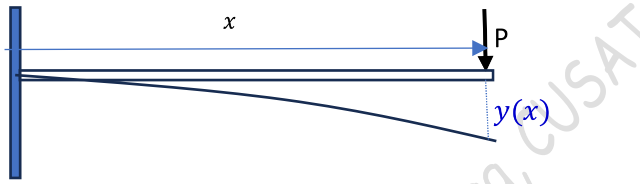
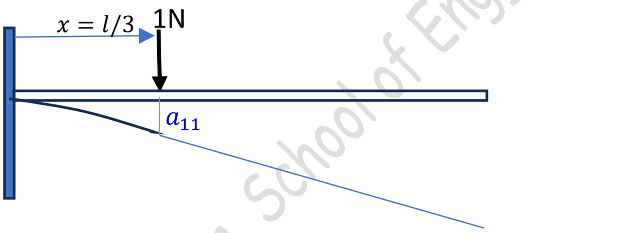
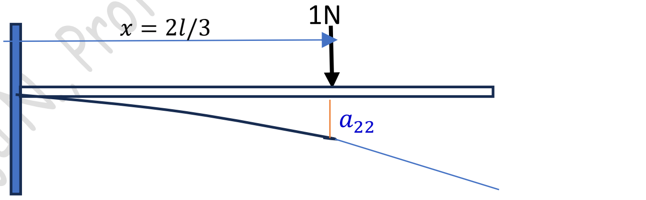
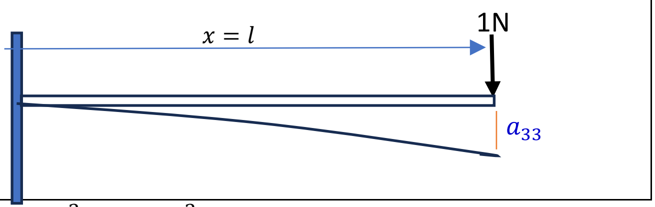
$$\frac{1}{{\omega_1}^2} \simeq a_{11}m_1 + a_{22}m_2 + a_{33}m_3 = \frac{3}{256}\frac{l^3}{EI}m_1 + \frac{1}{48}\frac{l^3}{EI}m_2 + \frac{3}{256}\frac{l^3}{EI}m_1 = \frac{ml^3}{EI}\left(\frac{3}{256} + \frac{1}{48} + \frac{3}{256}\right)$$

$$\frac{1}{{\omega_1}^2} = 0.04427\frac{ml^3}{EI}$$

$${\omega_1}^2 = 22.588\frac{EI}{ml^3}$$

$$\omega_1 = 4.753\sqrt{\frac{EI}{ml^3}}$$

Example 2 Determine the fundamental frequency of the cantilever beam using Dunkerly's formula. Assume $m_1 = m_2 = m_3$.	

The deflection of the cantilever beam under a load P at a distance x from the support $$y(x) = \frac{Px^3}{3EI}$$	
Flexibility influence coefficient $\boldsymbol{a_{11}}$ is the deflection at location 1 due to a unit load at location 1. $$\boldsymbol{a_{11}} = \frac{Px^3}{3EI} = \frac{(1N)\left(\frac{l}{3}\right)^3}{3EI} = \frac{l^3}{81EI}$$	
Flexibility influence coefficient $\boldsymbol{a_{22}}$ is the deflection at location 2 due to a unit load at location 2. $$\boldsymbol{a_{11}} = \frac{Px^3}{3EI} = \frac{(1N)\left(\frac{2l}{3}\right)^3}{3EI} = \frac{8l^3}{81EI}$$	
Flexibility influence coefficient $\boldsymbol{a_{33}}$ is the deflection at location 3 due to a unit load at location 3. $$\boldsymbol{a_{11}} = \frac{Px^3}{3EI} = \frac{(1N)(l)^3}{3EI} = \frac{l^3}{3EI}$$	

$$\frac{1}{{\omega_1}^2} \simeq a_{11}m_1 + a_{22}m_2 + a_{33}m_3 = \frac{l^3}{81EI}m_1 + \frac{8l^3}{81EI}m_2 + \frac{l^3}{3EI}m_1 = \frac{ml^3}{EI}\left(\frac{1}{81} + \frac{8}{81} + \frac{1}{3}\right)$$

$$\frac{1}{{\omega_1}^2} = 0.444\frac{ml^3}{EI}$$

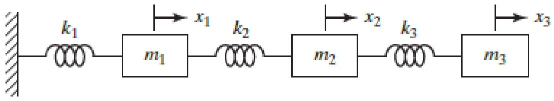
$${\omega_1}^2 = 2.25\frac{EI}{ml^3}$$

$$\omega_1 = 1.5\sqrt{\frac{EI}{ml^3}}$$

Example 3

Find the natural frequency of the given three degree of freedom system using Dunkerley's method.

Assume $m_1 = m_2 = m_3;\ \ and\ k_1 = k_2 = k_3$



Recall the values of the flexibility influence coefficients calculated for this system earlier

$$a_{11} = \frac{1}{k_1} = \frac{1}{k}$$

$$a_{22} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{2}{k}$$

$$a_{33} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} = \frac{3}{k}$$

$$\frac{1}{{\omega_1}^2} \simeq a_{11}m_1 + a_{22}m_2 + a_{33}m_3 = \frac{m}{k} + \frac{2m}{k} + \frac{3m}{k} = \frac{6m}{k}$$

$${\omega_1}^2 = \frac{1}{6}\frac{k}{m} = 0.167\frac{k}{m}$$

$$\omega_1 = 0.408\sqrt{\frac{k}{m}}$$

Note that the exact value of the fundamental frequency of this system was earlier found out to be equal to $\omega_1 = 0.445\sqrt{\frac{k}{m}}$. The Dunkerley's formula gives a fundamental frequency value lower than the exact value.

2. Rayleigh's Method

It is already discussed in the chapter of influence coefficients that for an n-degree-of freedom system the potential energy can be written in matrix form as

$$V = \frac{1}{2}\vec{x}^T\ [k]\vec{x}$$

and kinetic energy in the matrix form as,

$$T = \frac{1}{2}\vec{\dot{x}}^T\ [m]\vec{\dot{x}}$$

Assuming a harmonic solution $\{x\}$=$\{X\}\sin(\omega t + \phi)$ where $\{X\}$ is the modal vectors and ω is the natural frequency.

$$\{\dot{x}\}=\{X\}\omega\sin(\omega t + \phi)$$

The maximum value of the potential energy, $V_{max} = \frac{1}{2}\{X\}^T\ [k]\{X\}$.

(The potential energy is maximum when $\{x\}$ is maximum and it is equal to $\{X\}$ when $\sin(\omega t + \phi) = 1$)

The maximum value of the kinetic energy, $T_{max} = \frac{1}{2}\omega\{X\}^T\ [m]\{X\}\omega = \frac{1}{2}\omega^2\{X\}^T\ [m]\{X\}$.

(The kinetic energy is maximum when $\{\dot{x}\}$ is maximum and it is equal to $\omega\{X\}$ when $\sin(\omega t + \phi) = 1$)

For a conservative system (undamped), there is no energy loss.

$$T_{max} = V_{max}$$

$$\frac{1}{2}\omega^2\{X\}^T\,[m]\{X\} = \frac{1}{2}\{X\}^T\,[k]\{X\}$$

$$\omega^2 = \frac{\{X\}^T\,[k]\{X\}}{\{X\}^T\,[m]\{X\}}$$

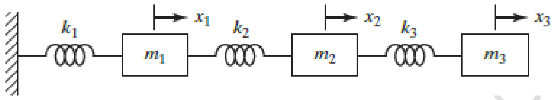
The right hand sid of the equation is known as Rayleigh's quotient.

Example 4

Find the natural frequency of the given three degree of freedom system using Dunkerley's method.

Assume $m_1 = m_2 = m_3;\ \ and\ k_1 = k_2 = k_3$

and a modal vector $\{X\} = \begin{Bmatrix}1\\2\\3\end{Bmatrix}$



Recall the exact solution of this 3-degree of freedom system.

$$[m] = \begin{bmatrix}m & 0 & 0\\0 & m & 0\\0 & 0 & m\end{bmatrix}$$

$$[k] = \begin{bmatrix}2k & -k & 0\\-k & 2k & -k\\0 & -k & k\end{bmatrix}$$

For $\omega_1 = 0.445\sqrt{\frac{k}{m}}$,the normal modes or modal vector is $\{X\} = \begin{Bmatrix}1\\1.802\\2.247\end{Bmatrix}$

Using Rayleigh's quotient,

$$\omega^2 = \frac{\{X\}^T\,[k]\{X\}}{\{X\}^T\,[m]\{X\}}$$

$$\{X\}^T\,[m]\{X\} = [1 \quad 2 \quad 3]\begin{bmatrix}m & 0 & 0\\0 & m & 0\\0 & 0 & m\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} = [m \quad 2m \quad 3m]\begin{bmatrix}1\\2\\3\end{bmatrix} = 14m$$

$$\{X\}^T\,[k]\{X\} = [1 \quad 2 \quad 3]\begin{bmatrix}2k & -k & 0\\-k & 2k & -k\\0 & -k & k\end{bmatrix}\begin{bmatrix}1\\2\\3\end{bmatrix} = [0 \quad 0 \quad k]\begin{bmatrix}1\\2\\3\end{bmatrix} = 3k$$

$$\omega^2 = \frac{\{X\}^T\,[k]\{X\}}{\{X\}^T\,[m]\{X\}} = \frac{3k}{14m} = 0.214\frac{k}{m}$$

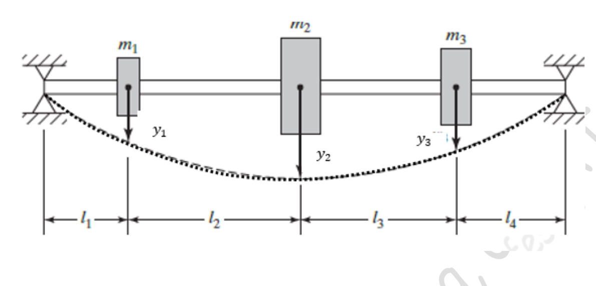
$$\omega = 0.463\sqrt{\frac{k}{m}}$$

Rayleigh's method gives a fundamental frequency higher than the exact solution.

3. Rayleigh's Method for beams and shafts

Although the Rayleigh's method described above is applicable to all discrete systems, another equation can be derived for finding out the fundamental frequency of transverse vibration of beams and shafts with pulleys and gears. Here the static deflection curve is approximated as the dynamic deflection curve

Consider a shaft carrying three pulleys of masses m_1, m_2, m_3 as shown. The static deflection of the shaft at the locations of the masses are y_1, y_2, y_3 respectively. The potential energy of the system is the strain energy of the deflected shaft which is equal to the work done by the static loads $m_1 g, m_2 g$, and $m_3 g$.



The maximum potential energy of the system,

$$V_{max} = \frac{1}{2}(m_1 g.y_1 + m_2 g.y_2 + m_3 g.y_3)$$

Assuming harmonic oscillation $y_i \sin \omega t$ for the masses $(i = 1, 2, and\ 3)$, the velocity will be $y_i \omega \cos \omega t$ and the maximum velocity will be $y_i \omega$.

The maximum kinetic energy of the system,

$$T_{max} = \frac{1}{2}(m_1.\omega^2 {y_1}^2 + m_2.\omega^2 {y_2}^2 + m_3.\omega^2 {y_3}^2) = \frac{1}{2}.\omega^2(m_1 {y_1}^2 + m_2 {y_2}^2 + m_3 {y_3}^2)$$

For a conservative (or undamped) system,

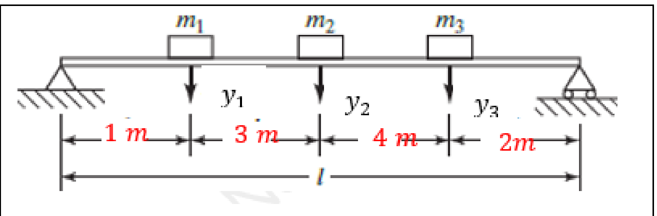
$$T_{max} = V_{max}$$

$$\frac{1}{2}.\omega^2(m_1 {y_1}^2 + m_2 {y_2}^2 + m_3 {y_3}^2) = \frac{1}{2}(m_1 g.y_1 + m_2 g.y_2 + m_3 g.y_3)$$

$$.\boldsymbol{\omega^2} = \frac{\boldsymbol{g(m_1 y_1 + m_2\, y_2 + m_3 y_3)}}{\boldsymbol{(m_1 {y_1}^2 + m_2 {y_2}^2 + m_3 {y_3}^2)}}$$

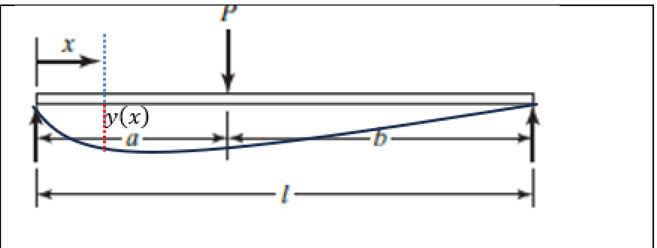
Example 5

Determine the fundamental frequency of the simply supported beam using Dunkerly's formula. Assume $m_1 = 20\ kg, m_2 = 50\ kg, m_3 = 40\ kg$.



The deflection of a simply supported beam at a location x from one support when a load P is acting at a distance b from the other support is given by

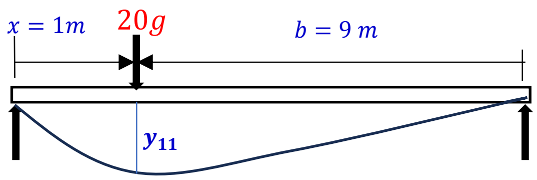
$$y(x) = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2); \quad 0 \le x \le a$$



The deflection at location 1 due to a load $m_1 g$ at location 1 is $\boldsymbol{y_{11}}$

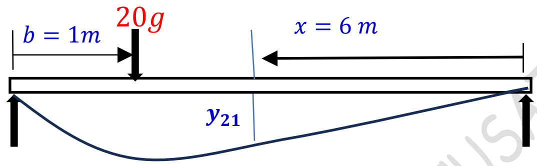
$$y_{11} = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2)$$

$$= \frac{(20 \times 9.81)(9)(1)(10^2 - 9^2 - 1^2)}{6EI \times 10} = \frac{529.74}{EI}$$



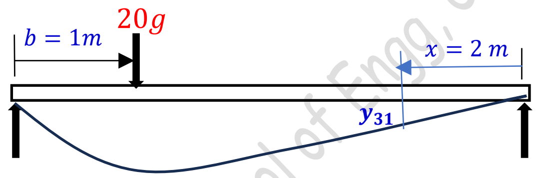
The deflection at location 2 due to a load $m_1 g$ at location 1 is $\boldsymbol{y_{21}}$

$$y_{21} = \frac{(20 \times 9.81)(1)(6)(10^2 - 1^2 - 6^2)}{6EI \times 10} = \frac{1236.06}{EI}$$



The deflection at location 3 due to a load $m_1 g$ at location 1 is $\boldsymbol{y_{31}}$

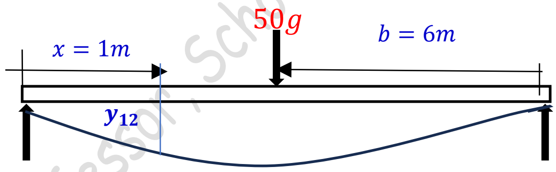
$$y_{31} = \frac{(20 \times 9.81)(1)(2)(10^2 - 1^2 - 2^2)}{6EI \times 10} = \frac{621.3}{EI}$$



The deflection at location 1 due to a load $m_2 g$ at location 2 is $\boldsymbol{y_{12}}$

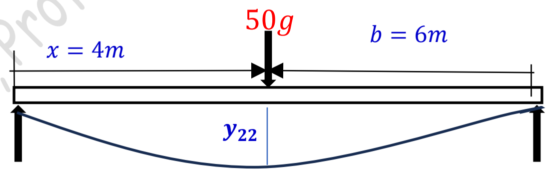
$$y_{12} = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2)$$

$$= \frac{(50 \times 9.81)(6)(1)(10^2 - 6^2 - 1^2)}{6EI \times 10} = \frac{3090.15}{EI}$$



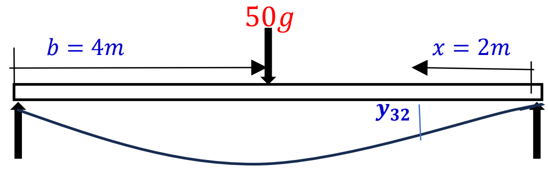
The deflection at location 2 due to a load $m_2 g$ at location 2 is $\boldsymbol{y_{22}}$

$$y_{22} = \frac{(50 \times 9.81)(6)(4)(10^2 - 6^2 - 4^2)}{6EI \times 10} = \frac{9417.6}{EI}$$



The deflection at location 3 due to a load $m_2 g$ at location 2 is $\boldsymbol{y_{32}}$

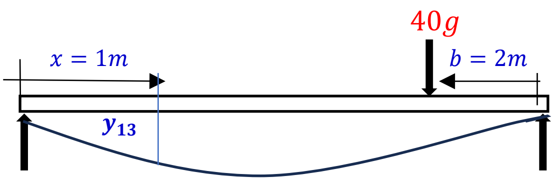
$$y_{32} = \frac{(50 \times 9.81)(4)(2)(10^2 - 4^2 - 2^2)}{6EI \times 10} = \frac{5232}{EI}$$



The deflection at location 1 due to a load $m_3 g$ at location 3 is $\boldsymbol{y_{13}}$

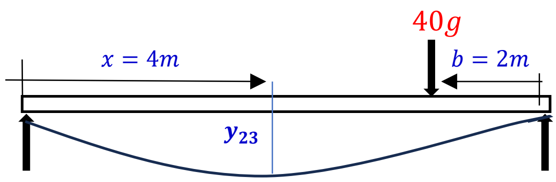
$$y_{13} = \frac{Pbx}{6EIl}(l^2 - b^2 - x^2)$$

$$= \frac{(40 \times 9.81)(2)(1)(10^2 - 2^2 - 1^2)}{6EI \times 10} = \frac{1242.6}{EI}$$



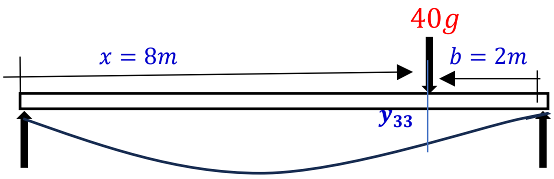
The deflection at location 2 due to a load $m_3 g$ at location 3 is $\boldsymbol{y_{23}}$

$$y_{23} = \frac{(40 \times 9.81)(2)(4)(10^2 - 2^2 - 4^2)}{6EI \times 10} = \frac{4185.6}{EI}$$



The deflection at location 3 due to a load $m_3 g$ at location 3 is $\boldsymbol{y_{33}}$

$$y_{33} = \frac{(40 \times 9.81)(2)(8)(10^2 - 2^2 - 8^2)}{6EI \times 10} = \frac{3348.48}{EI}$$



Therefore, the net deflection at location 1 due to all loads

$$y_1 = y_{11} + y_{12} + y_{13} = \frac{529.74}{EI} + \frac{3090.15}{EI} + \frac{1242.6}{EI} = \frac{4866.49}{EI}$$

the net deflection at location 2 due to all loads

$$y_2 = y_{21} + y_{22} + y_{23} = \frac{1236.06}{EI} + \frac{9417.6}{EI} + \frac{4185.6}{EI} = \frac{14839.26}{EI}$$

he net deflection at location 3 due to all loads

$$y_3 = y_{31} + y_{32} + y_{33} = \frac{621.3}{EI} + \frac{5232}{EI} + \frac{3348.48}{EI} = \frac{9201.78}{EI}$$

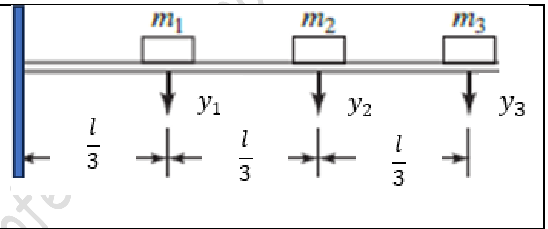
$$.\,\omega^2 = \frac{g(m_1y_1 + m_2y_2 + m_3y_3)}{(m_1{y_1}^2 + m_2{y_2}^2 + m_3{y_3}^2)} = \frac{9.81\left(20 \times \frac{4866.49}{EI} + 50 \times \frac{14839.26}{EI} + 40 \times \frac{9201.78}{EI}\right)}{20 \times \left(\frac{4866.49}{EI}\right)^2 + 50 \times \left(\frac{14839.26}{EI}\right)^2 + 40 \times \left(\frac{9201.78}{EI}\right)^2}$$

$$= \frac{\frac{9.81}{EI}}{\left(\frac{1}{EI}\right)^2}\left(\frac{20 \times 4866.49 + 50 \times 14839.26 + 40 \times 9201.78}{20 \times 4866.49^2 + 50 \times 14839.26^2 + 40 \times 9201.78^2}\right) = \frac{11844240.84}{1.487075 \times 10^{10}} EI = 0.000795\, EI$$

$$\omega = 0.0282\sqrt{EI}$$

Example 6

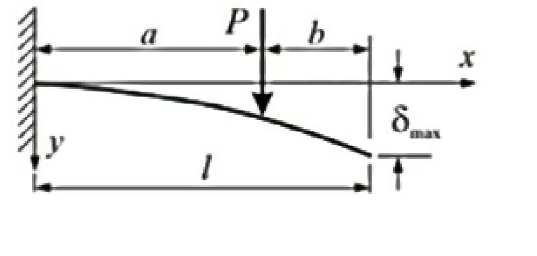
Determine the fundamental frequency of the cantilever beam using Rayleigh's method. Assume $m_1 = m_2 = m_3$.



The delection of a cantilver beam at a distance x from the support due to a point load P at a distance a from the support is given by

$$y = \frac{Px^2}{6EI}(3a - x) for\ 0 < x < a$$

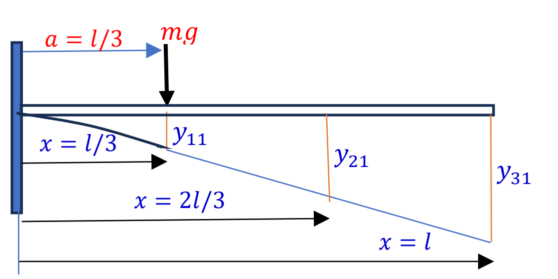
$$y = \frac{Pa^2}{6EI}(3x - a) for\ a < x < l$$



y_{11} is the deflection at location 1 due to a load mg at location 1.

$$y = \frac{Px^2}{6EI}(3a - x) for\ 0 < x < a\ ;\ a = \frac{l}{3}, and\ x = \frac{l}{3}$$

$$y_{11} = \frac{(mg)\left(\frac{l}{3}\right)^2\left[3\left(\frac{l}{3}\right) - \frac{l}{3}\right]}{6EI} = \frac{(mg)2\left(\frac{l}{3}\right)^3}{6EI} = \frac{2mgl^3}{162EI}$$



y_{21} is the deflection at location 2 due to a load mg at location 1.

$$y = \frac{Pa^2}{6EI}(3x - a) for\ a < x < l\ :\ a = \frac{l}{3}, and\ x = \frac{2l}{3}$$

$$y_{21} = \frac{(mg)\left(\frac{l}{3}\right)^2\left[3\left(\frac{2l}{3}\right) - \frac{l}{3}\right]}{6EI} = \frac{(mg)5\left(\frac{l}{3}\right)^3}{6EI} = \frac{5mgl^3}{162EI}$$

y_{31} is the deflection at location 3 due to a load mg at location 1.

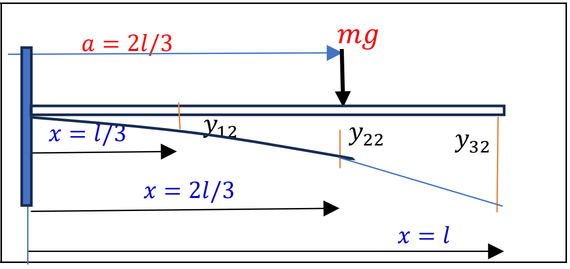
$$y = \frac{Pa^2}{6EI}(3x - a) for\ a < x < l\ :\ a = \frac{l}{3}, and\ x = l$$

$$y_{31} = \frac{(mg)\left(\frac{l}{3}\right)^2\left[3(l) - \frac{l}{3}\right]}{6EI} = \frac{(mg)2\left(\frac{l}{3}\right)^3}{6EI} = \frac{8mgl^3}{162EI}$$

$\boldsymbol{y_{12}}$ is the deflection at location 1 due to a load mg at location 2.

$$y = \frac{Px^2}{6EI}(3a - x) for\ 0 < x < a\,;\ \ a = \frac{2l}{3}\,, and\ x = \frac{l}{3}$$

$$\boldsymbol{y_{12}} = \frac{(mg)\left(\frac{l}{3}\right)^2\left[3\left(\frac{2l}{3}\right) - \frac{l}{3}\right]}{6EI} = \frac{(mg)5\left(\frac{l}{3}\right)^3}{6EI} = \frac{5mgl^3}{162EI}$$



$\boldsymbol{y_{22}}$ is the deflection at location 2 due to a load mg at location 2.

$$y = \frac{Pa^2}{6EI}(3x - a) for\ a < x < l\ :\ \ a = \frac{2l}{3}\,, and\ x = \frac{2l}{3}$$

$$\boldsymbol{y_{22}} = \frac{(mg)\left(\frac{2l}{3}\right)^2\left[3\left(\frac{2l}{3}\right) - \frac{2l}{3}\right]}{6EI} = \frac{(mg)2\left(\frac{2l}{3}\right)^3}{6EI} = \frac{16mgl^3}{162EI}$$

$\boldsymbol{y_{32}}$ is the deflection at location 3 due to a load mg at location 2.

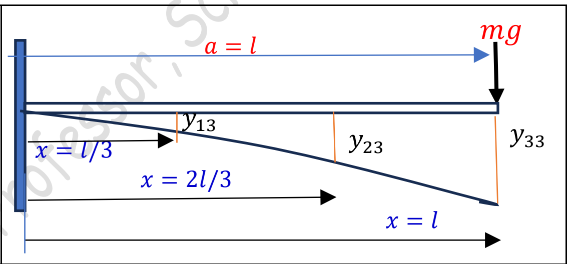
$$y = \frac{Pa^2}{6EI}(3x - a) for\ a < x < l\ :\ \ a = \frac{2l}{3}\,, and\ x = l$$

$$\boldsymbol{y_{32}} = \frac{(mg)\left(\frac{2l}{3}\right)^2\left[3(l) - \frac{2l}{3}\right]}{6EI} = \frac{(mg)\frac{28}{27}(l)^3}{6EI} = \frac{28mgl^3}{162EI}$$

$\boldsymbol{y_{13}}$ is the deflection at location 1 due to a load mg at location 3.

$$y = \frac{Px^2}{6EI}(3a - x) for\ 0 < x < a;\ \ a = l\,, and\ x = \frac{l}{3}$$

$$\boldsymbol{y_{13}} = \frac{(mg)\left(\frac{l}{3}\right)^2\left[3(l) - \frac{l}{3}\right]}{6EI} = \frac{(mg)8\left(\frac{l}{3}\right)^3}{6EI} = \frac{8mgl^3}{162EI}$$



$\boldsymbol{y_{23}}$ is the deflection at location 2 due to a load mg at location 3.

$$y = \frac{Px^2}{6EI}(3a - x) for\ 0 < x < a\ :\ \ a = l\,, and\ x = \frac{2l}{3}$$

$$\boldsymbol{y_{23}} = \frac{(mg)\left(\frac{2l}{3}\right)^2\left[3(l) - \frac{2l}{3}\right]}{6EI} = \frac{(mg)\frac{28}{27}(l)^3}{6EI} = \frac{28mgl^3}{162EI}$$

$\boldsymbol{y_{33}}$ is the deflection at location 3 due to a load mg at location 3.

$$y = \frac{Px^2}{6EI}(3a - x) for\ 0 < x < a\ :\ \ a = l\,, and\ x = l$$

$$\boldsymbol{y_{32}} = \frac{(mg)(l)^2[3(l) - l]}{6EI} = \frac{(mg)2(l)^3}{6EI} = \frac{54mgl^3}{162EI}$$

The net deflection at location 1, $y_1 = y_{11} + y_{12} + y_{13} = \frac{2mgl^3}{162EI} + \frac{5mgl^3}{162EI} + \frac{8mgl^3}{162EI} = \frac{15mgl^3}{162EI}$

The net deflection at location 2, $y_2 = y_{21} + y_{22} + y_{23} = \frac{5mgl^3}{162EI} + \frac{16mgl^3}{162EI} + \frac{28mgl^3}{162EI} = \frac{49mgl^3}{162EI}$

The net deflection at location 3, $y_3 = y_{31} + y_{32} + y_{33} = \frac{8mgl^3}{162EI} + \frac{28mgl^3}{162EI} + \frac{54mgl^3}{162EI} = \frac{90mgl^3}{162EI}$

$$.\omega^2 = \frac{g(m_1y_1 + m_2y_2 + m_3y_3)}{(m_1{y_1}^2 + m_2{y_2}^2 + m_3{y_3}^2)} = \frac{g\left(m \times \frac{15mgl^3}{162EI} + m \times \frac{49mgl^3}{162EI} + m \times \frac{90mgl^3}{162EI}\right)}{m \times \left(\frac{15mgl^3}{162EI}\right)^2 + m \times \left(\frac{49mgl^3}{162EI}\right)^2 + m \times \left(\frac{90mgl^3}{162EI}\right)^2}$$

$$= \frac{g\left(\frac{mgl^3}{162EI}\right)}{\left(\frac{mgl^3}{162EI}\right)^2}\left(\frac{15 + 49 + 90}{15^2 + 49^2 + 90^2}\right) = g\left(\frac{162EI}{mgl^3}\right) = 2.326\frac{EI}{ml^3}$$

$$\omega_1 = 1.525\sqrt{\frac{EI}{ml^3}}$$

Note that since the load acting at locations 1,2, and 3 are the same ($= mg$), the deflections satisfy Maxwell's Reciprocity Theorem.

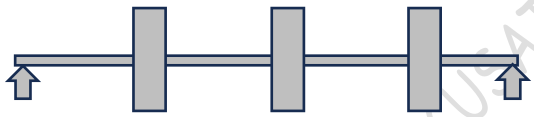
$$y_{12} = y_{21}; \quad y_{13} = y_{31}; \quad y_{32} = y_{23}$$

Also note that the fundamental frequency obtained by Rayleigh's method is slightly igher than that obtained by Dunkerley's method.

Exercise 1

Determine the fundamental frequency of the simply supported beam of 4m length carrying three pulleys of mass 50 kg each with equal spacing. Use Dunkerley's method and Rayleigh's Method.

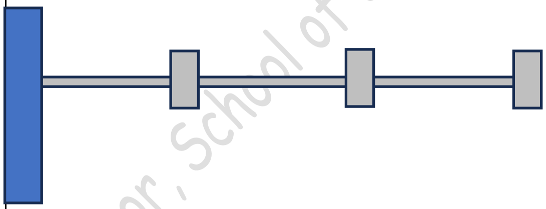
$$E = 200\ GPa, \qquad I = 8.5 \times 10^{-6}\ m^4$$



Exercise 2

Determine the fundamental frequency of the cantilever beam of 3m length carrying three pulleys of mass 50 kg each with equal spacing. Use Dunkerley's method and Rayleigh's Method.

$$E = 200\ GPa, \qquad I = 8.5 \times 10^{-6}\ m^4$$



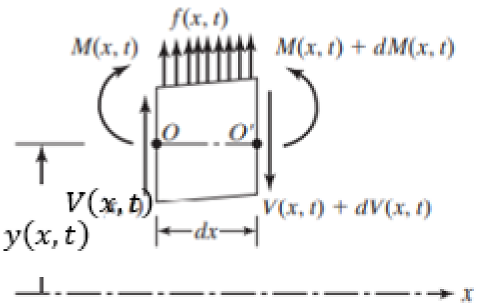
4. Rayleigh's method for continuous beams

Consider the freebody diagram of the beam. The elemental beam has a mass of $dm = \rho A\, dx$. The lateral deflection of the beam element is $y(x,t)$. Assuming a harmonic motion for the beam,

$$y(x,t) = Y(x)\sin(\omega t + \varphi)$$

Velocity, $\dot{y}(x,t) = Y(x)\omega\cos(\omega t + \varphi)$

The kinetic energy of the element, $dT = \frac{1}{2}.dm.(\dot{y})^2$



Total kinetic energy of the beam,

$$T = \int_0^l dT = \int_0^l \frac{1}{2}.dm.(\dot{y})^2 = \frac{1}{2}\int_0^l (Y\omega\cos\omega t)^2 \rho A\, dx$$

The kinetic energy will be maximum when velcity is maximum, $\dot{y}_{max} = Y(x)\omega$

$$T_{max} = \frac{1}{2}\omega^2 \int_0^l Y(x)^2 \rho A\, dx$$

The strain energy of the beam,

$$V = \frac{1}{2}\int_0^l M\, d\theta$$

Where M is the bending moment and θ is the slope of the deflected beam.

From the deflection equation of the beam,

$$EI\frac{\partial^2 y}{\partial x^2} = M \quad and \quad \theta = \frac{\partial y}{\partial x} \implies d\theta = \frac{\partial^2 y}{\partial x^2}\, dx$$

$$\therefore V = \frac{1}{2}\int_0^l \left(EI\,\frac{\partial^2 y}{\partial x^2}\right)\left(\frac{\partial^2 y}{\partial x^2}\,dx\right) = \frac{1}{2}\int_0^l EI\left(\frac{\partial^2 y}{\partial x^2}\right)^2 dx$$

Since the maximum value of y is $Y(x)$,

$$V_{max} = \frac{1}{2}\int_0^l EI\left(\frac{\partial^2 Y(x)}{\partial x^2}\right)^2 dx$$

By equating V_{max} and T_{max}, we get the Rayleigh's quotient

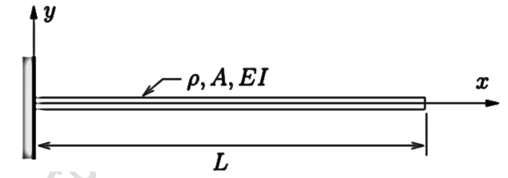
$$\frac{1}{2}\omega^2\int_0^l Y(x)^2\rho A\,dx = \frac{1}{2}\int_0^l EI\left(\frac{\partial^2 Y(x)}{\partial x^2}\right)^2 dx$$

$$\omega^2 = \frac{\int_0^l EI\left(\frac{\partial^2 Y(x)}{\partial x^2}\right)^2 dx}{\int_0^l Y(x)^2\rho A\,dx}$$

Example 7

Find the fundamental frequency of transverse vibration of a contilever beam of length L and constant ρ, A, EI with an assumed mode shape of

$$Y(x) = \frac{1}{6EI}(3Lx^2 - x^3)$$



Note that the given $Y(x) = \frac{1}{6EI}(3Lx^2 - x^3)$ is the equation for the deflected shape of the cantilever when a unit point load is acting at its end.

$$\frac{\partial Y(x)}{\partial x} = \frac{1}{6EI}(6Lx - 3x^2)$$

$$\frac{\partial^2 Y(x)}{\partial x^2} = \frac{1}{6EI}(6L - 6x) = \frac{(L-x)}{EI}$$

$$\omega^2 = \frac{\int_0^L EI\left[\frac{(L-x)}{EI}\right]^2 dx}{\int_0^L \left[\frac{1}{6EI}(3Lx^2 - x^3)\right]^2 \rho A\,dx} = \frac{36EI\int_0^L [L^2 - 2Lx + x^2]dx}{\rho A\int_0^L [9L^2x^4 - 6Lx^5 + x^6]\,dx}$$

$$= \frac{36EI\left[L^2x - 2L\frac{x^2}{2} + \frac{x^3}{3}\right]_0^L}{\rho A\left[9L^2\frac{x^5}{5} - 6L\frac{x^6}{6} + \frac{x^7}{7}\right]_0^L} = \frac{36EI\left(\frac{L^3}{3}\right)}{\rho A\left(\frac{33L^7}{35}\right)} = 12.7273\frac{EI}{\rho AL^4}$$

$$\omega = 3.5675\sqrt{\frac{EI}{\rho AL^4}}$$

Compare this value with the exact fundamental frequency derived earlier $\omega_1 = 3.516015\sqrt{\frac{EI}{\rho AL^4}}$

Example 8

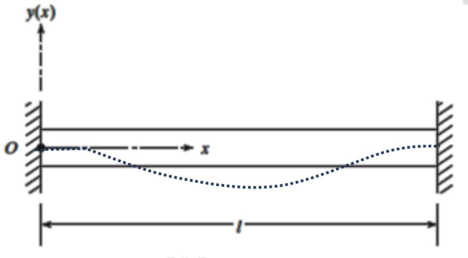
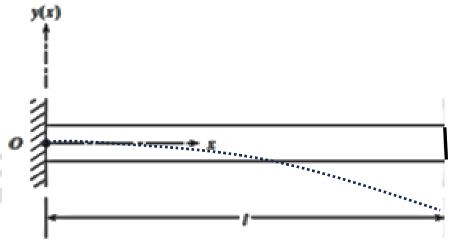
Find the fundamental frequency of transverse vibration of a simply supported beam of length L and constant ρ, A, EI with an assumed mode shape of $Y(x) = C\sin\frac{\pi x}{L}$

$$\frac{\partial Y(x)}{\partial x} = C\left(\frac{\pi}{L}\right)\cos\frac{\pi x}{L}$$

$$\frac{\partial^2 Y(x)}{\partial x^2} = -C\left(\frac{\pi}{L}\right)^2 \sin\frac{\pi x}{L}$$

$$\omega^2 = \frac{\int_0^L EI\left[-C\left(\frac{\pi}{L}\right)^2 \sin\frac{\pi x}{L}\right]^2 dx}{\int_0^L \left[C\sin\frac{\pi x}{L}\right]^2 \rho A\, dx} = \frac{EI\left(\frac{\pi}{L}\right)^4 \int_0^L \left[\sin^2\left(\frac{\pi x}{L}\right)\right] dx}{\rho A \int_0^L \left[\sin^2\left(\frac{\pi x}{L}\right)\right] dx} = \pi^4 \frac{EI}{\rho A L^4}$$

$$\omega = \pi^2 \sqrt{\frac{EI}{\rho A L^4}}$$

Exercise 3 Determine the fundamental frequency of transverse vibration for the fixed -fixed beam of uniform cross section $20\,cm \times 30cm$, length $3m$, density $8000\,kg/cm^3$, and modulus of elasticity $E = 200\,GPa$. Assume the shape of the deflected beam as $Y(x) = C\left(1 - \cos\frac{2\pi x}{L}\right)$	
Exercise 3 Determine the fundamental frequency of transverse vibration for the cantilever beam of uniform cross section $20\,cm \times 30cm$, length $3m$, density $8000\,kg/cm^3$, and modulus of elasticity $E = 200\,GPa$. Assume the shape of the deflected beam as $Y(x) = C\left(1 - \cos\frac{\pi x}{2L}\right)$	

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